

Frequency-Coverage Limits in Assessing Gravity Signals from Oceanic Mass Redistribution

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Abstract

Oceanic mass redistribution produces direct gravitational attraction and, for surface loads, elastic/potential and height responses over timescales from seconds to months. We develop an evidence-bounded process–distance–instrument framework with explicit source geometry, observation duration, spectral coverage, and instrument-specific noise. The primary six-process production ensemble contains 1,446 records of direct radial gravity; elastic and height components are retained separately where evaluated and are not implied for every process branch. None passes a predeclared 90% signal-energy coverage gate for the three admitted vertical-gravity literature anchors: tide, storm surge, eddy and internal-wave signals have no overlapping band, whereas tsunami and landslide records have only partial coverage. We therefore withhold SNR classifications rather than extrapolating instrument curves. A separate open-data Helgoland calculation reproduces the published BSH gravity-to-height ratio within 15.6%. The median lower edge required for 90% energy coverage ranges from 3.89×10^{-8} Hz for the eddy scenario to 2.63×10^{-4} Hz for tsunami packets, factors of 1.3×10^4 and 1.9 below the lowest reviewed 5×10^{-4} Hz curve edge. The framework converts a withheld classification into quantitative frequency requirements while keeping source geometry and mass compensation as separate physical limits.

1 Introduction

Ocean tides, non-tidal ocean loading and storm surges already contribute to terrestrial gravity and displacement observations [2, 3, 29, 30]. Superconducting and atom-interferometric gravimeters reach high precision, but their sensitivity, stability, self-noise and site noise are different quantities with different frequency support [1, 23, 24, 20, 21].

Environmental Newtonian-noise studies provide the broader theory of gravity fluctuations from moving density fields [11, 28]. Work for gravitational-wave facilities often reports strain-equivalent noise from seismic or atmospheric sources [6, 4, 10]; those results are not vertical-gravimeter noise curves. Long terrestrial records also require removal of tides and other background components, and different procedures can leave residuals or distort target signals [27].

Simple comparison of a source peak amplitude with one quoted sensitivity value therefore cannot establish detectability. The source spectrum must overlap a measured noise spectrum, the observable and units must agree, and enough signal energy must lie inside the reported band before an SNR is authorized.

Here we ask that coverage question under one auditable convention for six representative process families. The contribution is a cross-process direct- radial-gravity framework, an explicit coverage gate, quantitative lower-band- edge requirements, mass-compensation tests and an open Helgoland model validation. The gravity kernels, elastic loading theory and instrument concepts are prior work, and no universal detectable/undetectable classification is claimed.

2 Related physical observables

Direct attraction is the Newtonian acceleration at a specified observation point. A surface load also deforms the Earth and changes its potential; a terrestrial gravimeter can additionally respond to station-height change. In general these contributions must be combined only after their reference frame, sign and Green-function conventions have been validated [8, 7, 18]. The six-process production ensemble in this paper contains direct radial gravity, not that total terrestrial observable.

Gravity gradient $T_{ij} = \partial g_i / \partial x_j$ has units of s^{-2} and a different instrument class [19, 25]. Gravitational-wave equivalent strain is another derived observable tied to detector geometry and a transfer function. Gradient and strain foundations are retained as supplemental context; neither is compared with the vertical-gravity curves used here.

Figure 1 fixes the observable boundary used throughout the paper. It is a provenance diagram, not a claim that every process branch has a validated total-observable calculation.

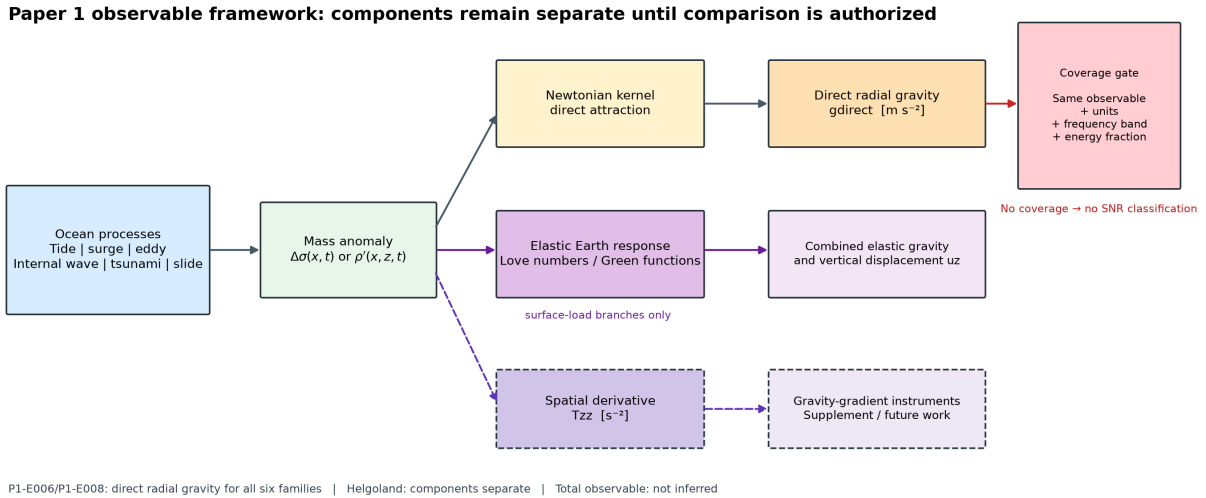


Figure 1: Observable and evidence framework. Ocean mass anomalies feed the direct radial-gravity calculation; surface loads may additionally require elastic/potential and station-height terms before a total terrestrial observable can be formed. Gravity gradient is a separate supplemental observable. The spectral-energy coverage gate must pass before a noise-based classification is authorized. Arrows describe calculation dependencies, not completed branches for all six processes.

3 Methods

3.1 Observables and sign conventions

All kernels use SI units. For a surface mass anomaly $\Delta\sigma$,

$$\Delta \mathbf{g}_{\text{direct}}(\mathbf{r}, t) = G \int_A \Delta\sigma(\mathbf{x}, t) \frac{\mathbf{x} - \mathbf{r}}{|\mathbf{x} - \mathbf{r}|^3} dA. \quad (1)$$

For a volume-density anomaly ρ' , the analogous integral is evaluated over volume. The gravity-gradient convention is $T_{ij} = \partial g_i / \partial x_{\text{observation},j}$. Direct attraction, combined elastic gravity and vertical displacement are stored separately. The LoadDef CE gravity term combines deformation and potential response according to its source convention and is never represented as direct attraction.

3.2 Earth, grid and coastal geometry

Reference calculations include local planar, mean-radius spherical, and WGS 84 ellipsoidal geometry. Spherical and ellipsoidal cell areas are integrated in their native surface coordinates. Missing ocean values, hard masks, and partial coastal-cell fractions are explicit. Grid refinement and domain truncation are tested before a dataset is admitted to a result.

For idealized compact surface anomalies evaluated against terrestrial instruments, source cells and the station lie on a spherical Earth. Distance is the great-circle surface standoff from the represented three-standard-deviation source support. Compact anomalies translate parallel to an idealized coastline, preventing a source cell from coinciding with the station. A rejected local-plane near-field calculation is retained in the experiment preregistration history: its midpoint quadrature exceeded the continuous-sheet amplitude bound and did not enter the registered output.

3.3 Ocean-process models

The six process families are summarized in Table 1. Surface and compensated three-dimensional variants are retained only where physically supported. Engineering fixtures used for software validation are excluded from scientific uncertainty statements. Parameter values are linked to named events, catalogue means or published ranges. They define sensitivity designs, not occurrence probabilities or population percentiles. The mesoscale-eddy Gaussian branch uses catalogue-scale amplitude, radius and propagation speed from Chelton et al. [5]; observed energetics and vertical structure show why SSH does not uniquely specify an interior density anomaly [17, 14, 9]. Internal-wave scales are anchored to field and satellite studies [15, 12, 31, 22], while tsunami and landslide branches retain named-source provenance [26, 16, 13]. Surface-wave packets and density dipoles are mass balanced before gravity integration.

Table 1: Primary direct-radial-gravity representations in P1-E006. Record counts include repeated distances, durations or paired sensitivity points and are not counts of independent natural events.

Process	Evidence/representation	Independent physical family	Records
Tide	1-m M2 spherical-disk transfer	one geometry diagnostic; 5 distances $\times$ 32 durations	160
Storm surge	gridded 2022 Helgoland BSH field	one named model case	1
Eddy	catalogue-mean translating Gaussian SSH	one representative mean case; 5 distances	5
Internal wave	compensated density dipole	one model family; 128 paired sensitivities $\times$ 5 distances	640
Tsunami	mass-balanced Gaussian packet	one packet family; 96 paired sensitivities $\times$ 5 distances	480
Landslide	Storegga conserved relocation	one named extreme family; 32 paired sensitivities $\times$ 5 distances	160

3.4 Spectra and detectability

For transient signals, expected matched-filter signal-to-noise ratio is

$$\rho^2 = 4 \int_{f_{\min}}^{f_{\max}} \frac{|\tilde{s}(f)|^2}{S_n(f)} df. \quad (2)$$

Fourier normalization, detrending, one-sided PSD, and observation duration are versioned. A classification is forbidden unless the instrument curve covers at least 90% of signal spectral energy. Sensitivity is evaluated separately at 50%, 75%, 90% and 95%; these are coverage fractions, not detection probabilities. Vertical gravity, gravity gradient, and gravitational-wave equivalent strain remain different observables and are never merged into one sensitivity curve. The admitted vertical-gravity entries are an iGrav quiet self-noise anchor over 10^{-3} – 10^{-1} Hz and short-term AQG-A01 and FG5#228 white-noise anchors over the source-reported 5×10^{-4} – 10^{-2} Hz plateau; their intervals are not extrapolated.

For a finite, constant-detrended signal, the coverage quantity is

$$C(f_{\text{low}}, f_{\text{high}}) = \frac{\int_{f_{\text{low}}}^{f_{\text{high}}} |\tilde{s}(f)|^2 df}{\int_{0^+}^{f_{\text{Nyquist}}} |\tilde{s}(f)|^2 df}. \quad (3)$$

The registered primary convention uses a rectangular window and trapezoidal integration of positive frequency Fourier amplitudes. An independent review audit also applies rectangular and periodic Hann window one-sided periodogram bin sums. The maximum admissible lower edge is the highest resolved frequency for which C reaches the requested fraction at a fixed upper edge.

3.5 Parameter design and uncertainty

Latin-hypercube and stratified designs are labelled as scenario envelopes unless every dimension has a source-supported probability distribution. Engineering ranges cannot be reported as probabilistic 5–95% intervals. Density-compensated models retain cancellation and far-field suppression rather than replacing them with equivalent uncompensated water columns. Source standoffs are 10, 30, 100, 300 and 1,000 km, and every coverage-limited outcome is retained.

4 Validation

Analytic and regression tests cover point masses, finite disks, rectangles, Gaussian sheets, volume cells, gradient symmetry and trace, WGS 84 area, mass conservation, grid convergence, domain truncation, and time/frequency normalization. Test counts and software-behaviour details are reported in the reproducibility supplement; the scientific validation below rests on analytic errors, published benchmarks and the independent Helgoland case rather than the number of tests.

The LoadDef v1.2.2 PREM/CE provider reproduces all 50 published benchmark rows used by the project for angular distance, radial displacement and combined elastic gravity [18]. Direct attraction is calculated independently; LoadDef’s Newtonian term is not added a second time.

The Helgoland model-only reproduction uses 242 BSH-HBMnoku files and 720 hourly fields conservatively remapped to a 1/8-degree WGS 84 grid. Direct attraction, combined elastic gravity and vertical displacement remain separate. After the registered sign conversion at the literature-comparison boundary, the computed gravity-to-height ratio is $-2.266 \text{ nm s}^{-2} \text{ mm}^{-1}$, compared with $-2.684 \text{ nm s}^{-2} \text{ mm}^{-1}$ reported by Voigt et al. [29]. The 15.6% difference passes the predeclared

20% tolerance for replacing SPOTL/Gutenberg–Bullen and a three-year tidal analysis with LoadDef PREM/CE and the registered 30-day harmonic fit.

The registered component audit quantifies, but does not over-interpret, this difference. The direct term has a fitted ratio of $-0.406 \text{ nm s}^{-2} \text{ mm}^{-1}$ and RMS and peak-to-peak amplitudes equal to 18.0% and 18.1% of the combined elastic term. Their diagnostic arithmetic sum is $-2.672 \text{ nm s}^{-2} \text{ mm}^{-1}$, 0.45% from the published ratio, but this is not an authorized comparison: the registered target is the combined elastic quantity and the source paper’s observable convention cannot be redefined post hoc. PREM/CE versus SPOTL/Gutenberg–Bullen Earth response and the 30-day fit versus three-year detiding remain confounded in the available computation, so no causal percentage is assigned to either source.

5 Results

Figure 2 compares process timescales with the admitted instrument-curve support. M2 tide and the M2 internal-wave branch lie near $2.24 \times 10^{-5} \text{ Hz}$. Storm-surge and eddy ranges are lower. The iGrav anchor begins at 10^{-3} Hz and the reviewed AQG/FG5 plateaus at $5 \times 10^{-4} \text{ Hz}$. Tsunami characteristic content crosses these edges, while the Storegga transition lies mainly below them; neither has 90% of signal energy inside a complete admitted band.

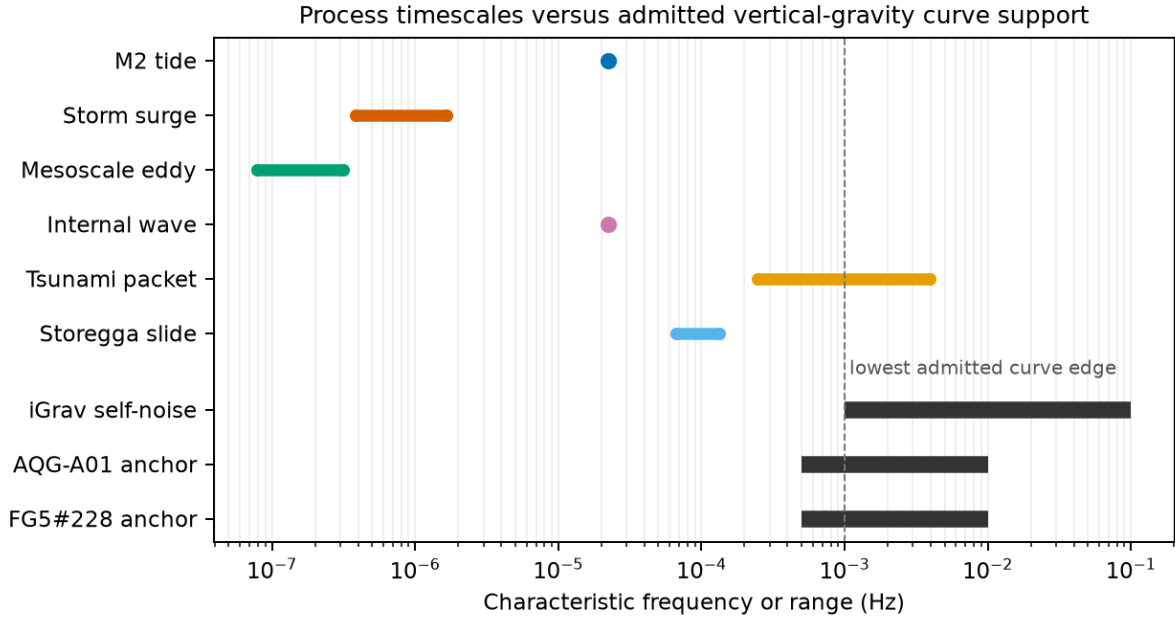


Figure 2: Characteristic frequencies of the direct-radial-gravity process signals and admitted vertical-gravity curve support. Points denote fixed constituent frequencies and horizontal bars denote sensitivity ranges. Curve limits are not extrapolated.

The resulting coverage matrix is retained as versioned Supplementary Figure S1. All 806 long-period records have no covered spectral interval for any admitted curve. All 640 tsunami and Storegga records have some overlapping bins but fail the 90% energy criterion. Consequently no production record is labelled detectable or undetectable. This differs from a statement of zero SNR: available curve support is insufficient to make the comparison. In Supplementary Figure S1, “no

band” means fewer than two positive-frequency bins overlap a curve and “partial” means overlap is present but covers less than 90% of signal energy.

Figure 3 converts that withheld classification into an instrument-evidence requirement. At 90% coverage, median admissible lower edges are 3.89×10^{-8} Hz for the eddy, 7.72×10^{-7} Hz for the storm surge, 1.10×10^{-5} Hz for the internal wave, 2.15×10^{-5} Hz for the tide, 4.92×10^{-5} Hz for the landslide and 2.63×10^{-4} Hz for the tsunami. Relative to the lowest reviewed 5×10^{-4} Hz edge, the corresponding gaps are approximately 12,900, 648, 45, 23, 10 and 1.9.

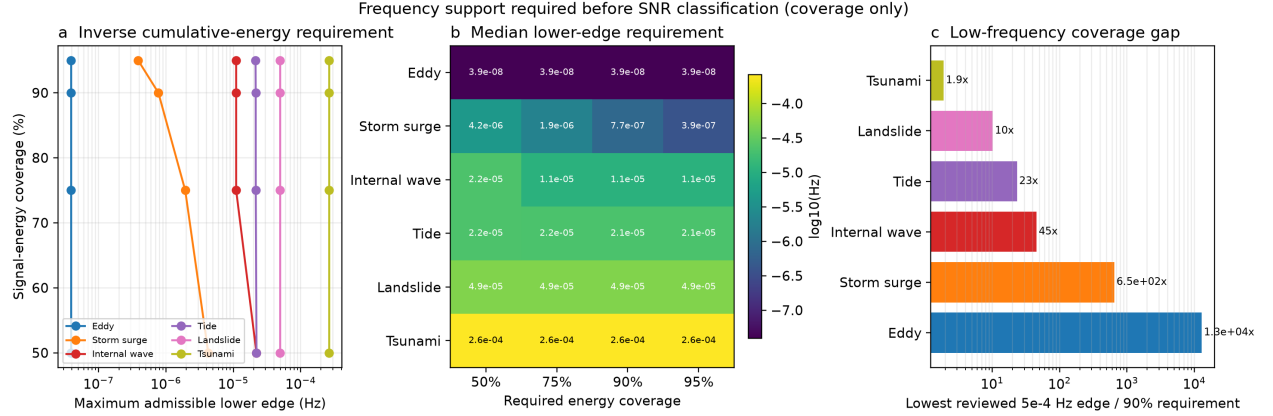


Figure 3: Direct-radial-gravity frequency requirements. (a) Median inverse cumulative-energy requirement at 50%, 75%, 90% and 95% coverage; connecting segments guide the eye. (b) Median maximum admissible lower band edge. (c) Gap between the lowest reviewed 5×10^{-4} Hz lower edge and the 90% requirement. These are coverage quantities, not SNRs or detection probabilities.

Direct radial-gravity amplitudes decay with spherical surface standoff (Figure 4). The Storegga named-event branch has the largest primary-branch envelope, with median peak amplitude decreasing from about $9.5 \times 10^{-9} \text{ m s}^{-2}$ at 10 km to $1.0 \times 10^{-10} \text{ m s}^{-2}$ at 1,000 km. The compensated tsunami envelope decreases from approximately $3.1\text{--}6.5 \times 10^{-10}$ to $4.9\text{--}12.7 \times 10^{-12} \text{ m s}^{-2}$ across the same standoffs. The internal-wave dipole has an off-axis maximum near 100 km, illustrating why amplitude need not decrease monotonically inside a compensated source’s near field.

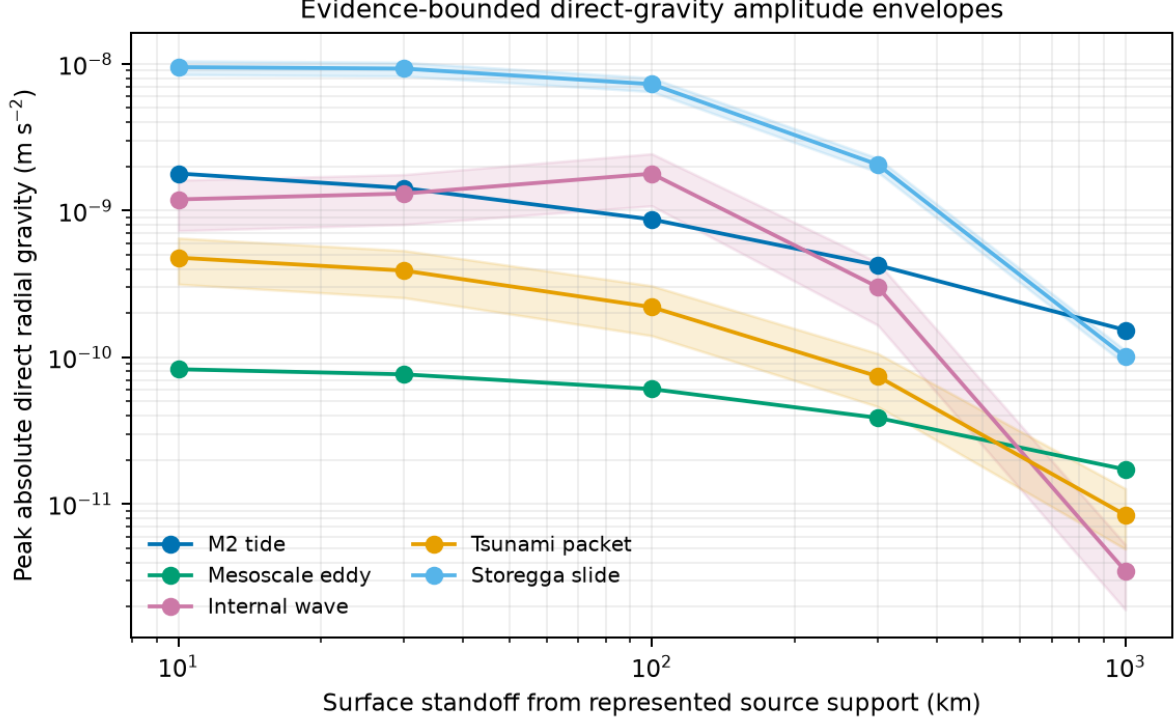


Figure 4: Peak absolute direct radial gravity versus surface standoff. Lines show the median sensitivity design and shading spans the sampled minimum and maximum; shading is not a probability interval.

Versioned Supplementary Figure S2 separates the density/displacement, wave-packet and slide-parameter envelopes. Every tsunami crest and trough pair cancels signed surface mass to numerical tolerance, and every slide record relocates rather than creates solid mass. Its bands are design extrema, not confidence or credible intervals.

A separate registered structural-closure experiment maps the 3.0×10^{15} J Tohoku propagated-energy estimate (6% uncertainty) reported by Tang et al. [26] to the same idealized mass-balanced Gaussian packet. Under the declared linear-long-wave convention $E = \rho g \int \eta^2 dA$, the 300–400 km separation sensitivity gives a central crest amplitude of 4.359 m and an energy-uncertainty range of 4.226–4.488 m. This branch is deliberately separate from the 1.0–1.8 m deep-ocean amplitude-normalized branch; their difference measures normalization/model-form sensitivity and is not combined as a probability interval.

Figure 5 closes the main results by placing conservation and the Helgoland component audit in one view. It preserves the distinction between physical closure, a validated elastic comparison and an informative but unauthorized arithmetic diagnostic.

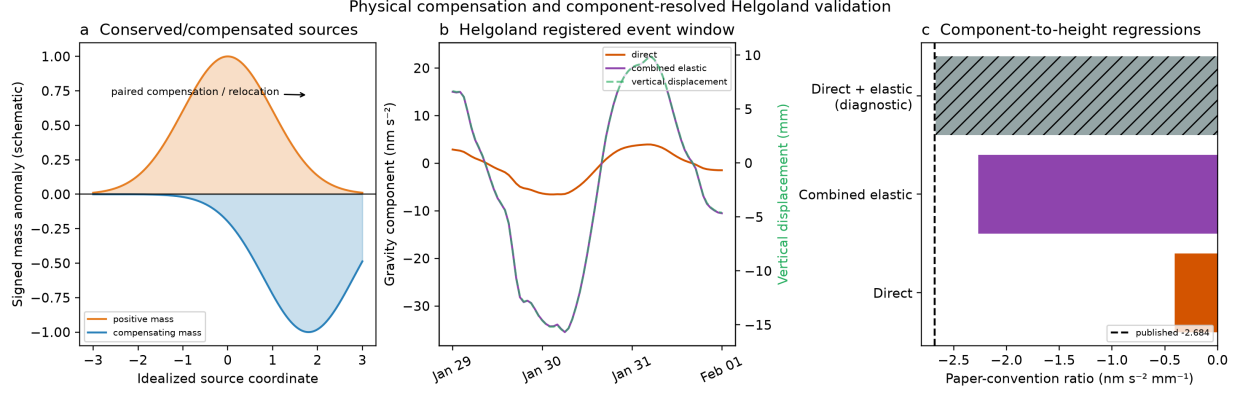


Figure 5: Physical structure and Helgoland validation. (a) Schematic only: compensated wave/density anomalies and a conserved slide relocation suppress the far-field monopole relative to an uncompensated load. (b) Separately stored Helgoland direct gravity, combined elastic gravity and vertical displacement. (c) Fitted gravity-to-height ratios. Only the combined elastic bar has the registered comparison to the published value. The hatched direct-plus-elastic bar is an arithmetic diagnostic and is not an authorized alternative observable or acceptance test.

6 Discussion

The principal result is a boundary between classified, coverage-limited, and physically suppressed signals. Compensated internal structures can decay much faster than surface-load estimates, while long-period processes may be strong but unclassifiable when empirical noise curves do not cover their frequency band. An amplitude–noise intersection is not a sufficient detectability test: observation duration and spectral energy coverage must be evaluated first.

The absence of a production SNR classification is therefore informative. It identifies a data requirement—site and instrument noise curves extending into the process band—rather than an invitation to extend short-term white-noise anchors by assumption. Environmental Newtonian-noise calculations expressed as gravitational-wave strain are likewise not interchangeable with gravimeter self-noise [4].

The threshold analysis distinguishes robust conclusions from finite-record sensitivity. All six median 90% requirements lie below 5×10^{-4} Hz, so the coverage-limited result does not depend on choosing 90% rather than 50% or 95%. The storm-surge requirement nevertheless shifts by about one decade between 50% and 95%, whereas several short transient designs remain on the same resolved Fourier bin. Future field PSDs should therefore extend below the process-specific requirement and use records long enough to resolve its threshold dependence.

The review audit reconstructed all 1,446 signals using the source-reported AQG and FG5 intervals and repeated the 50–95% calculation with rectangular and periodic Hann periodogram bin sums. Every record retains its no-band or partial-band status. Median 90% requirements remain below the lowest reviewed edge for every process and convention. The largest convention effect is the internal-wave rectangular periodogram result, 2.20×10^{-5} Hz, twice the registered trapezoidal value but still a factor of 23 below 5×10^{-4} Hz. These checks establish robustness to the numerical convention, not natural population uncertainty.

The Helgoland reproduction supplies an independent positive validation of the elastic loading scale but not of observation/model correlation. The reported 85 nm s^{-2} and -34 mm extrema are observations, not BSH model acceptance targets. Reproducing the reported correlation and RMS

reduction would require authenticated iGrav047 data and a paper-equivalent processed 2022 GNSS series.

7 Conclusions

The current public instrument anchors do not support a decision-grade cross-process detectability map: no production record reaches the required signal-energy coverage. This result does not imply absent physical signals or universal non-detectability. It establishes the frequency support that future site-specific noise measurements must provide: median 90% lower edges span 3.89×10^{-8} to 2.63×10^{-4} Hz across the six scenarios. The direct-gravity scenario suite quantifies how geometry, distance and mass compensation alter the source signal. Total terrestrial observables require separately validated elastic/potential and height components and are not inferred from the direct atlas alone.

8 Data and code availability

Code, compact experiment metadata, checksums, figure inputs and manuscript sources are maintained in the public project repository. Bulk BSH NetCDF and GNSS RINEX inputs remain on the compute data volume and are represented by source queries, inventories and checksums. Restricted gravimeter data and large gridded products remain outside Git and are represented by versioned manifests, access conditions, time ranges, and checksums.

9 Limitations

The production atlas contains one primary branch per process family, plus the energy-normalized tsunami sensitivity. Remaining structural branches are explicit scientific exclusions, not hidden omissions or zero-effect assumptions. The eddy design cannot simultaneously hold peak SSH, radius, and positive mass fixed across Gaussian and compact quadratic profiles; the compensated eddy lacks an SSH-to-density-profile mapping. Continuous mode-1 and free-surface internal-wave branches lack a boundary-conditioned vertical profile and coupling coefficient. Mediterranean/Storfjorden slide evidence does not jointly close amplitude and duration, and the generated-water-wave branch lacks an event-specific generation/entrainment transfer model.

The admitted instrument curves are literature anchors rather than complete site-specific environmental noise models. The iGrav curve estimates quiet self-noise, AQG and FG5 entries are short-term anchors, and none authorizes low-frequency extrapolation. Thus the matrix establishes a coverage boundary, not universal instrument performance. Haikou observation reproduction remains author-restricted; this does not affect the six-process ensemble. A PEGS simulator is unrelated to Paper 1. The repository release uses collective project authorship; journal submission still requires named author order, affiliations, declarations and corresponding-author approval.

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